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FINNISH SOCIAL SCIENCE
DATA ARCHIVE

Finnish Social Science Data Archive

Data Steward Training Course

March 3, 2025

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FSD Mission statement

*Our mission is to **preserve** data collected to study **Finnish** society, people and cultural phenomena in the **long term**. We **disseminate** data to reusers reliably and efficiently through our modern digital services.*

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Finnish Social Science Data Archive (FSD)

Tampere University

- Founded in 1999, separate unit of Tampere University
- National research data infrastructure
- CoreTrustSeal certified data repository
- Main user groups: researchers, students and teachers in universities
- Free-of-charge services for users
- Finnish Service Provider for CESSDA ERIC

CORE TRUST SEAL

cessda

FSD's Aila data portal's key numbers

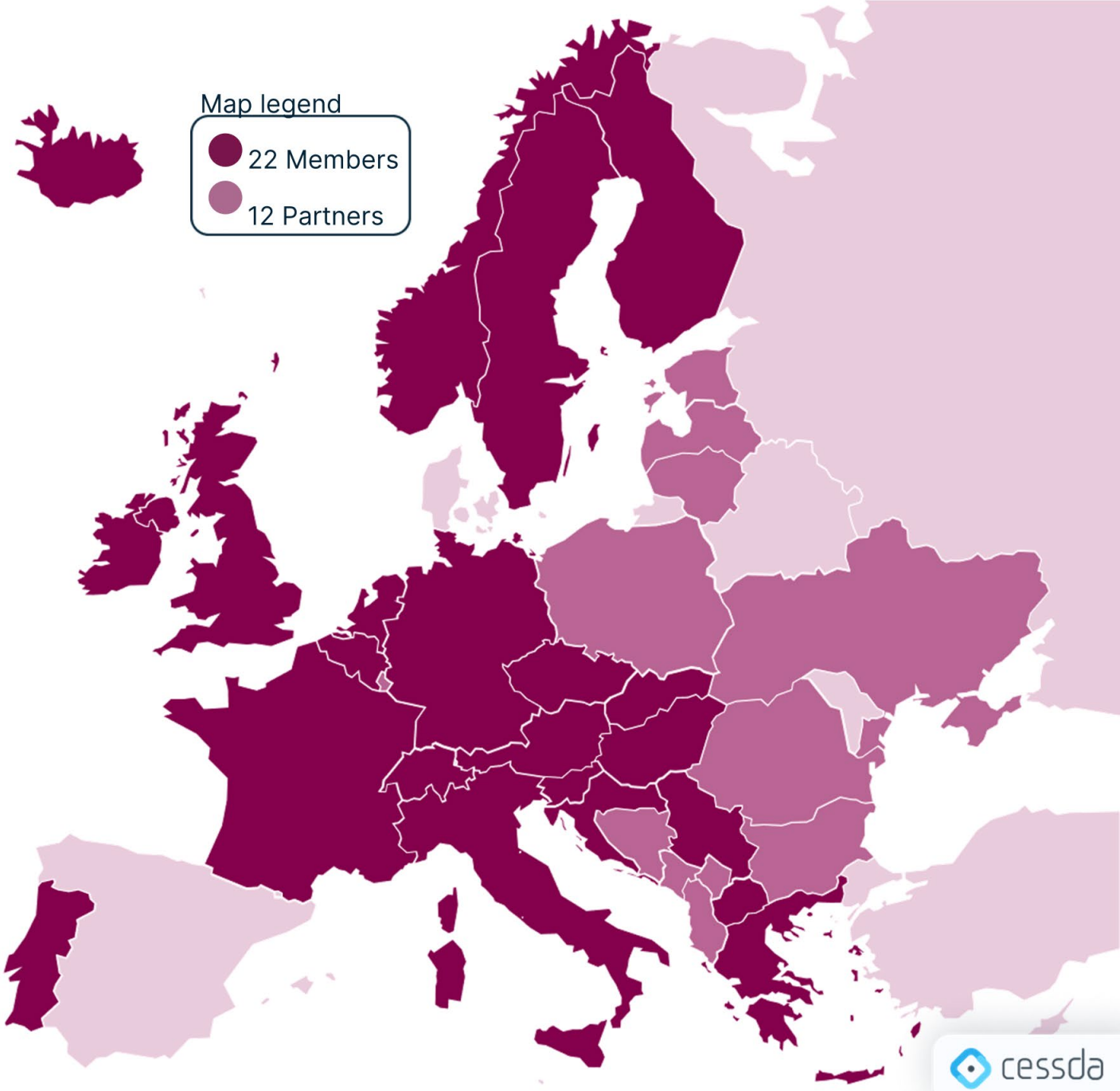
- ~6500 registered users (2024)
- ~2050 data sets
- ~6500 access applications (2024)
- ~6200 publications citing data

CESSDA

Is the Consortium of European Social Science Data Archives (Service Providers)
A true European research infrastructure:
1 out of 24 ERICs in Europe

Mission

To provide a sustainable research infrastructure that enables the research community to conduct high-quality research in the social sciences
to contribute to effective solutions to the major challenges facing society today.



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FSD's Main Functions

Findable
Accessible
Interoperable
Reusable

- Data archiving
  - Ingesting, processing and curating data (soc. sci and related)
  - Metadata production (descriptive metadata in DDI)
  - Digital preservation and persistent identifiers & data citation
  - Related software development, open APIs
- Delivering data to users (researchers, students, teachers) and metadata to machines
- Providing information services related to data and data management
  - Data Management Guidelines
  - Research Methods Guidebook (in Finnish)
- Promoting data reuse and open science, comparative research
- Participating in national and international data services collaboration
  - For example, FSD is a member of ICPSR and our data is findable in the CESSDA Data Catalogue, actively involved in developing the DDI metadata standard.

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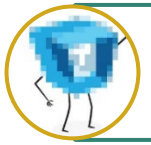
Research data at FSD

Both quantitative and qualitative

Data files + metadata = dataset



Archived data is edited: with preservation and data reusability as goals

Only digital data

No publications – i.e. not a library

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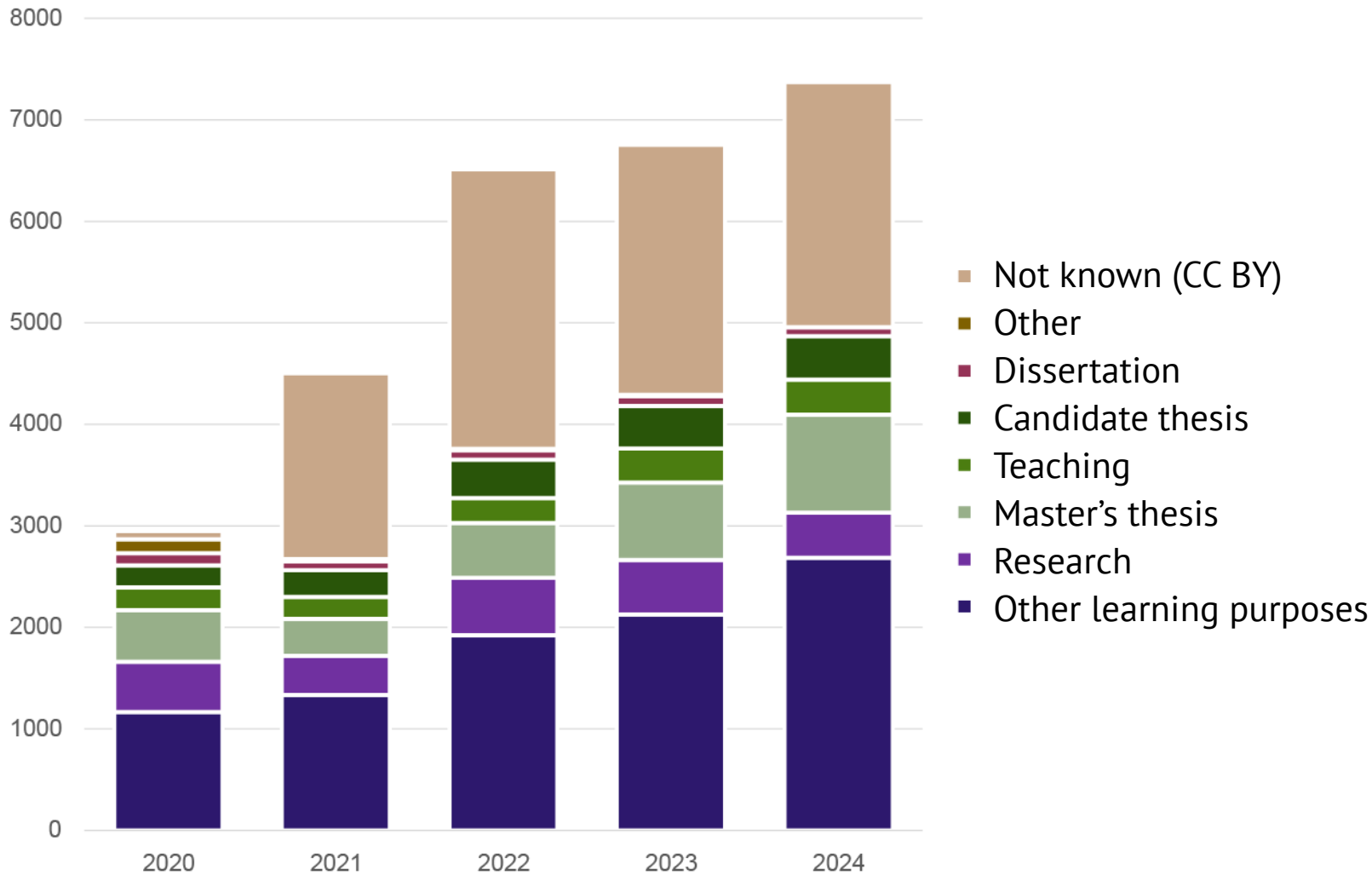
Data downloads 2020-20024



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Services

- Aila Data Service (deposit/discovery/dissemination)
- Data Management Guidelines
- Research Methods Guidebook (in Finnish)
- Information Service and Data Curation
- Penna – Textual data collection tool
- Pohtiva – Finnish Party Agenda Database
- FSD website, news, articles, blogs, guidance
- Longitudinal Metadata Portal (LODA)
  - the Finnish National Election Study series

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FSD's collaboration in the national context

Open science coordination

Finn-ARMA and data training coordination
(data management, data description)

EOSC Finland
RDA Finland

Digital preservation cooperation

GLAM sector collaboration

Platform cooperation

National Library and other bodies: Board memberships

Collaboration within Tampere University
(training, data support, digital preservation, IT, premises)

Ad hoc cooperation with national memory organisations and RPO's

FSD Advisory Board

- About 6-8 staff members take part in national collaboration, in total in ~20 different groups
- In addition: collaboration with users (via information services and archiving services) and with research groups (via projects e.g. ISSP, FIRIPO)

Exchange of knowledge and policies

- National collaborations
- Consortium of European Social Science Data Archives (CESSDA ERIC) and its Service Providers
  - European Open Science Cloud via CESSDA, projects and task forces
- ICPSR
- Standard bodies (e.g. DDI Alliance)
- Conferences (e.g. IASSIST, EDDI)
- Working groups and boards (e.g. CoreTrustSeal Board, RDA)
- National and international projects (e.g. EOSC-related)

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Standards, models and best practices at FSD

- Critical that our procedures are up-to-date, transparent and consistent with relevant standards, and our stakeholders trust us
- Standards in use:
  - OAIS and certification (CoreTrustSeal)
  - DDI and Controlled Vocabularies
  - Finnish national digital preservation service specifications
    - incl. METS, PREMIS
- FAIR principles, TRUST principles
- FSD's records management and archives formation plan
  - E.g. selection criteria, preservation planning, work processes, data security, continuity

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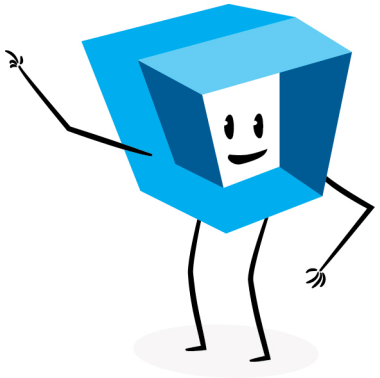
Who is FSD for?

- Researchers, teachers and students in higher education
  - As data depositors and data users/downloaders, or data collectors
  - As methods learners
  - Mostly in SSH but almost all disciplines covered
- Data stewards & repository staff, developers
  - Guidance, metadata and APIs
- Journalists, teachers in other schools
- Libraries, archives and museums
- Everyone
  - Information services, open metadata, methods, special collections (like Pohtiva),



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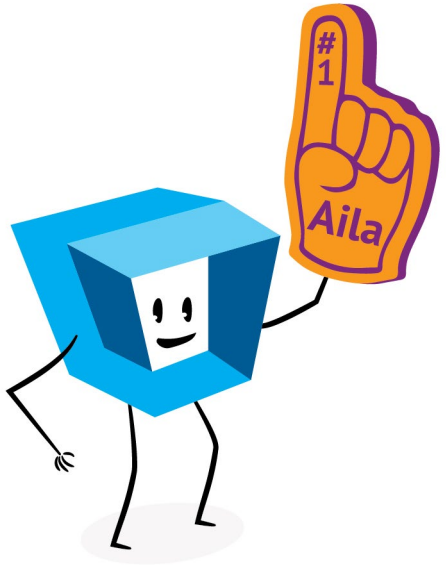
Why data reuse?

- Lots of new aspects to study, data rarely exhausted
  - New theories and methods create new possibilities
  - Support or compare findings
- Validity and reproducibility of results (FAIR)
- No need to bother research subjects repeatedly
- No need to handle personal data
- Saves time



Restricted access supports openness =>
data is as open as possible, as restricted as necessary

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Data stewardship at FSD

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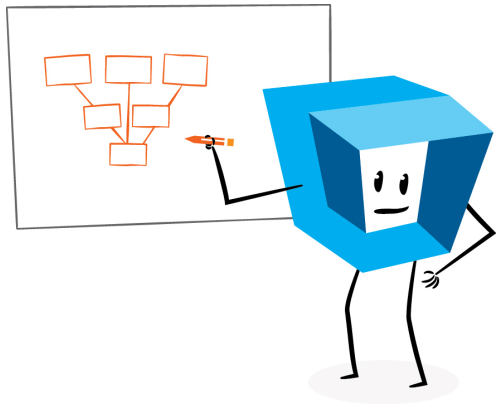
Working at FSD

- 20+ staff (15 full time, project staff varies)
- Information Specialists, Senior Specialists, Chief Specialists, Software Developers, Heads of Services, Project Specialists, Research Assistants
- Lots of inhouse software used to manage provenance and (meta)data
  - Aila portal and internal TIIPII information management system
  - Defined software stack and architecture for compliance and APIs
- Data processed in DDI XML, SPSS
  - Acrobat, Office and inhouse tools
  - Special software for qualitative data



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Data stewardship in practice at FSD

- Actively curate data deposited by the researchers (for the long-term and quality)
- Guide researchers to document, anonymise, clean and structure the data pre deposit
- Check / process / clean the data (formats)
- Produce descriptive variable level metadata for reuse (fi/en, DDI-C 2.5, vocabularies)
- Store, publish and share (or embargo) the data
  - Issue persistent identifiers, provide citations
  - Provide metadata through UIs and APIs
  - Enrich, migrate/convert data and metadata
- Collect information on publications based on the data
- Advice on and maintain data protection and security
  - Follow residual risk, minimization
- Special case: Textual data collection with Penna. Agreement with the researcher, managing collection, providing the qualidata set and archiving it

We do not
- approve Data Management Plans
- support active research phase
- carry out full anonymisation or data cleaning

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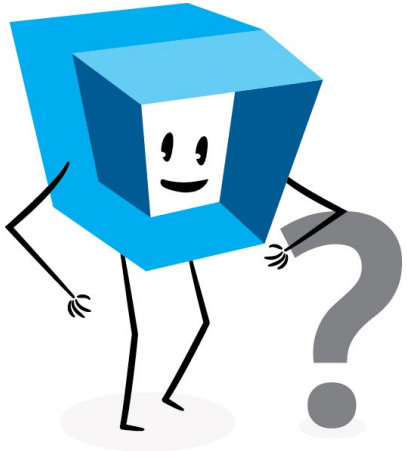
What do FSD data stewards need in the future?

- Understanding of the evolving designated community needs
- Metadata
- AI and IT tooling
- Subject specific questions but also generalist knowledge
- Ability to code and communicate
- Policy understanding (institutional, national, international)
- Look for competed funding
- Certification
- Customer service
- Networking



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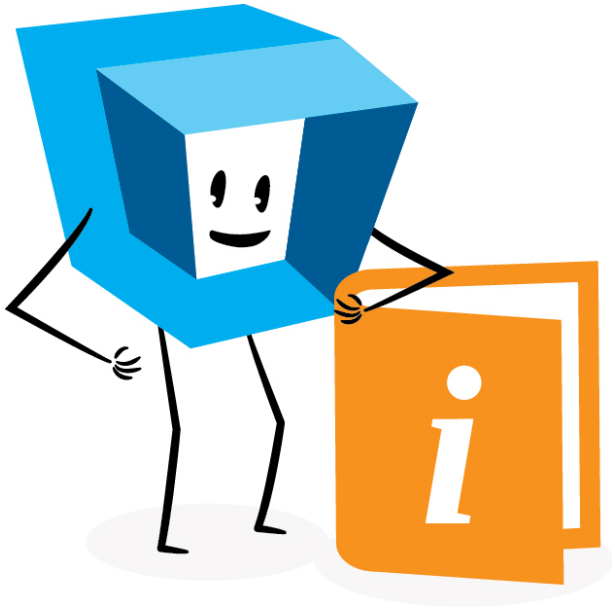
Data steward training at or offered by FSD

- Anonymisation training
  - Online guides
  - Tailored training
- Online Guides on Data lifecycle
- Deposit guide
  - Structuring data for reuse
- Responsible openness
  - GDPR, ethics, informing research participants
- Repository certification and FAIR
  - Project based, mentoring
- Train from the scratch. Until now!
  - Perhaps some of you will work at FSD some day!



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Is there still more time?
Because I have more slides!

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Aila

- Data Service Portal: data downloads but also deposit
  - Metadata is always available, even when the data is not open
- Aila is based on federated HAKA authentication
  - Students and staff from the Finnish universities and research institutions can register themselves using HAKA authentication
- Others apply for a user ID from FSD User Services
- Terms and conditions for data reuse vary:
  - (A) openly available for all users without registration
  - (B) available for research, teaching and study (1528 datasets)
  - (C) available for research only (including Master's, doctoral and University of Applied Sciences Master's theses)
  - (D) available only by permission from the data depositor/creator

Terms of use

- Registering to Aila requires you to follow the terms of use
- Keep the data safe and delete it after the use
- Remember that the right to use data is always personal
- If you notice mistakes etc. let us know
- Cite the data used in your research output
- Send us the citation to your work
- https://services.fsd.tuni.fi/docs/terms-of-use?lang=en

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SSH data & Open Science & Legal Aspects

- Data gathered from human subjects by surveying, interviewing, measuring, observing, or by the research subjects' own creation (written accounts, photographs, recordings...)
- Research Subjects should not be identifiable (GDPR & national law)
  - anonymisation
  - pseudonymisation
- Data ownership & licensing: data belongs to the depositors and may meet the threshold of originality
- Restricted access supports openness =>
  data is as open as possible, as restricted as necessary

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In addition to FAIR FSD helps with…

- Long-term digital preservation of data
  - FAIR requirements require that metadata and data are functional and readable in the future too. Burden off from researchers!
  - Data and metadata formats → understandability of data guaranteed
  - This also supports research repeatability / reproducibility

- Data protection and data security
  - Main expertise in FSD, noted also in the CoreTrustSeal certificate
  - For example, GDPR and anonymisation expertise, we support researchers already when planning the data collection

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Let’s discuss!

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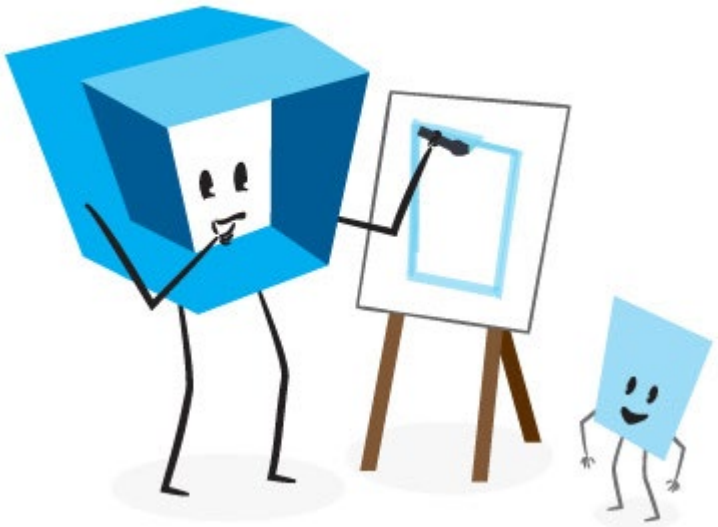

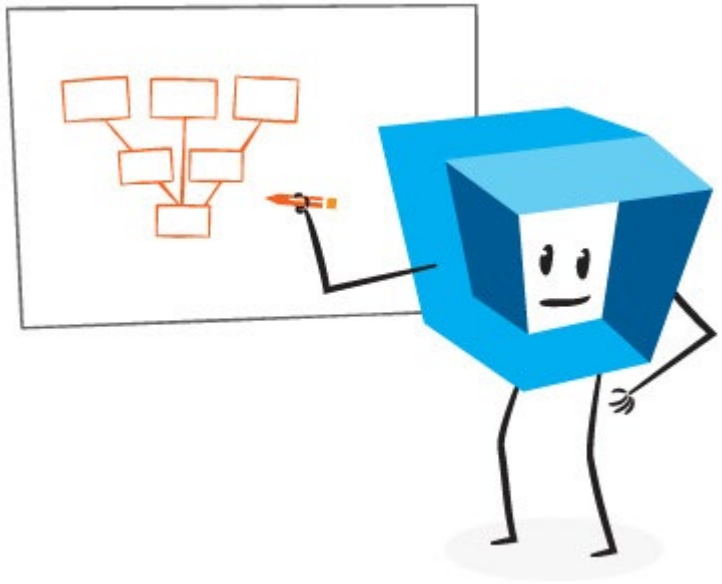

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